

PRIME: PSF Reconstruction and Identification for Multiple-source characterization Enhancement – application to Keck NIRC2 imager

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ABSTRACT

In order to enhance the scientific exploitation of adaptive optics (AO)-assisted observations, we investigate a novel hybrid concept to improve the parametric estimation of point spread function (PSF) called PSF Reconstruction and Identification for Multiple-source characterization Enhancement (PRIME). PRIME uses both focal and pupil-plane measurements to estimate jointly the model parameters related to the atmosphere [$C_n^2(h)$, seeing] and the AO system (e.g. optical gains and residual low-order errors). Photometry and astrometry are provided as by-products. The parametric model in use is flexible enough to be scaled with field location and wavelength, making it a proper choice for optimized on-axis and off-axis data-reduction across the spectrum. Here, we present the methodology and validate PRIME on engineering and binary Keck II telescope NIRC2 images. We also present applications of PSF model parameters retrieval using PRIME: (i) calibrate the PSF model for observations void of stars on the acquired images, i.e. optimize the PSF reconstruction process, (ii) update the AO error breakdown mutually constrained by the telemetry and the images in order to speculate on the origin of the missing error terms and evaluate their magnitude, and (iii) measure photometry and astrometry with an application to the triple system Gl569 images.

Key words: atmospheric effects – instrumentation: adaptive optics – methods: analytical – methods: data analysis – astrometry.

1 INTRODUCTION

Estimation of key science quantities, such as astrometry and photometry, are made possible by data-reduction techniques. For sources extraction or modelling, common pipelines are StarFinder (Diolaiti et al. 2000), SEXTRACTOR (Bertin & Arnouts 1996), or DAOPHOT (Stetson 1987). None the less, past analysis (Sheehy, McCrady & Graham 2006; Fritz et al. 2010; Schödel 2010; Yelda et al. 2010) has assessed that 50 percent of the astrometry and photometry error breakdown in the Galactic Center (GC) is induced by a problem of determination of the point spread function (PSF). A more recent study (Ascenso et al. 2015) has illustrated that photometry in the general case of globular clusters observations assisted by adaptive optics (AO) can be obtained at better than 1 percent level by providing StarFinder with the exact PSF, advocating for pushing the PSF characterization further to reach more accurate science estimates. One may rapidly realize that the

source confusion and measurement noise are important limitations to extract the PSF.

An alternative approach to identify the AO PSF is PSF reconstruction (PSF-R; Veran et al. 1997) that estimates the PSF from real-time AO control loop data, i.e. wavefront sensor (WFS) measurements and deformable mirror (DM) commands. Such a concept has already been demonstrated on-sky (Veran et al. 1997; Jolissaint, Ragland & Wizinowich 2015; Martin et al. 2016; Ragland et al. 2016, 2018) and gives astronomers access to a PSF model independently from any processing of the science image. PSF-R has the great advantage of being insensitive to the science detector noise and potential source confusion in the science image to estimate the PSF. Moreover, it offers a way to provide the PSF when no stars are present within the science field.

However, PSF-R lacks operational use so far because it requires a large amount of specific AO skills, especially to capture comprehensively AO performance and behaviour on-sky. For instance, in the case of AO-assisted observations performed at the W.M. Keck Observatory with a quad-cells WFS, the determination of WFS op-

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tical gains plays a major role in the PSF-R and is highly sensitive to seeing conditions and may vary unexpectedly (Ragland et al. 2016).

Moreover, the AO telemetry permits to reconstruct the PSF in the Guide Star (GS) direction only. To extrapolate the PSF across the field of view (FOV), one must model the spatial variations due to anisoplanatism (Fried 1982; Beltramo-Martin et al. 2018), which requires the knowledge of atmospheric parameters such as the $C_n^2(h)$. For single-conjugated AO (SCAO) systems that rely on a single Natural Guide Star (NGS) or a Laser Guide Star (LGS) plus an NGS to measure tip-tilt modes, the $C_n^2(h)$ cannot be measured from AO control loop data, which necessitates obtaining it from external instruments (Wilson 2002; Butterley, Wilson & Sarazin 2006; Osborn et al. 2018) pointing at a different sky direction than the telescope. Discrepancies at a level of 10 per cent up to 20 per cent on the $C_n^2(h)$ estimation compared to AO control loop data-based technique on multiple NGS/LGS systems have been already observed (Ono et al. 2016), which have an impact of the off-axis PSF model accuracy.

On top of that, the PSF is affected by field-dependent static aberrations that must be characterized by using long-exposure phase diversity (Mugnier et al. 2008) or focal-plane sharpening (Lamb et al. 2016) for instance, which potentially requires a large amount of telescope time (Sitarski et al. 2014). Finally, earlier studies have showed discrepancies between the reconstructed PSFs and the on-sky PSFs (Jolissaint et al. 2015; Ragland et al. 2016), which are attributed to the presence of instrumental effects, addressed at the system level (Ragland et al. 2016, 2018) and telemetry issues.

In this paper, we present PSF Reconstruction and Identification for Multiple-source characterization Enhancement (PRIME), which addresses the discrepancies in the post-processing stage. PRIME aims at optimizing the PSF-R process by mutually using pupil-plane (AO control loop data) and focal-plane (science images) information. By relying on a non-linear least-squared minimization, PRIME retrieves characteristics of both the atmosphere [seeing, $C_n^2(h)$] and the AO system (optical gains and residual low-order error) by adjusting those into the PSF forward model to match focal-plane observations. Thanks to the PSF-R framework, one may extrapolate the PSF model at any wavelength and sky directions by introducing anisoplanatism (Beltramo-Martin et al. 2018) after the PSF-R calibration operated. PRIME is actually combining PSF-R and iterative non-linear minimization algorithms to improve eventually the AO PSF estimation. In Section 2, we recall the PSF-R framework inherited from the work presented in Ragland et al. (2018) and Jolissaint et al. (2015) and explicit our strategy to optimize the process by adding the information coming from the focal plane. We highlight the key model parameters and validate the approach performance in simulation and analyse its sensibility with respect to the star magnitude.

In Section 3, we present a validation of PRIME using AO loop data acquired on the Keck II telescope as well as synchronous images obtained with the NIRC2 imager (Larkin et al. 2006). We reconstruct Strehl ratio (SR) and full width at half-maximum (FWHM), at 1 per cent level, as well as photometry and astrometry precision down to 0.005 mag and 100 μ s on $m_H = 14$ mag binaries.

Finally, Section 4 illustrates three different applications of PRIME beyond the PSF estimation: (i) PSF-R calibration in order to still provide a PSF from AO telemetry only when the focal-plane image does not contain any point source bright enough, (ii) update the error-breakdown that inherits from the PSF-R framework by constraining it regarding jointly the PSF morphology and the AO telemetry, (iii) measure relative photometry and astrometry in stellar fields as illustrated on the G1569 triple system.

2 PSF-R WITH PRIME

2.1 Forward reconstructed PSF model

PSF-R models the system optical transfer function (OTF) in closed loop from AO control loop data and the robustness and efficiency of this method is well established in the literature (Veran et al. 1997; Flicker 2008; Gilles et al. 2012; Jolissaint et al. 2015; Martin et al. 2016; Ragland et al. 2018). Our analysis focuses on the Keck SCAO system guided by either an NGS or an LGS and relies on the PSF-R concept implemented by Jolissaint et al. (2015).

We have identified a small number of key parameters whose estimation is one of the PSF-R limitation. We define μ the meta-parameters vector that gathers the $C_n^2(h)$ the atmospheric profile, high-order (HO) WFS and tip-tilt (TT) gains g_{dm} and g_{tt} and $\mathbf{a}_z$ a list of Zernike mode coefficients that allow to include an additional static phase. By assuming that the residual phase is stationary across the pupil, which is a reasonable hypothesis according to (Jolissaint et al. 2018), the OTF can be split into a product as follows:

$$\widehat{\text{OTF}}_e(\boldsymbol{\rho}/\lambda, \mu) = \text{OTF}_{\text{stat}}(\boldsymbol{\rho}/\lambda, \mathbf{a}_z) \mathbf{K}_{\perp}(\boldsymbol{\rho}/\lambda, r_0) \mathbf{K}_{tt}(g_{tt}) \mathbf{K}_{dm}(g_{dm}, \boldsymbol{\rho}/\lambda, g_{dm}) \mathbf{K}_{\text{alias}}(\boldsymbol{\rho}/\lambda, r_0) \mathbf{K}_{\text{aniso}}(\boldsymbol{\rho}/\lambda, C_n^2(h)) \quad (1)$$

where

(i) OTF_{stat} includes the pupil model and the calibrated static aberrations φ_{nepa} that contains non-common path aberrations (NCPA). For Keck observations, the calibration procedure is detailed in (Sitarski et al. 2014; Ragland et al. 2016; Witzel et al. 2016). On top of φ_{nepa} , we add n_z Zernike modes as follows:

$$\varphi_{\text{stat}}(\mathbf{r}, \mathbf{a}_z) = \varphi_{\text{nepa}}(\mathbf{r}) + \sum_{i=3}^{3+n_z} \mathbf{a}_z Z_i(\mathbf{r}), \quad (2)$$

where $\mathbf{r}$ the phase sample coordinates inside the pupil. Then, we calculate OTF_{stat} using

$$\text{OTF}_{\text{stat}}(\boldsymbol{\rho}/\lambda, \mathbf{a}_z) = \text{autocorr}(\mathcal{P} \exp(i\varphi_{\text{stat}}(\mathbf{r}))), \quad (3)$$

where $\text{autocorr}(x)$ is the autocorrelation function of x , $\mathcal{P}$ is the pupil function on which the static phase is defined, $\boldsymbol{\rho}$ is the separation vector within the pupil, and λ the wavelength.

Practically, we limit $n_z \leq 3$ to adjust only focus and astigmatism modes that are the likeliest present in the static aberrations and also the ones that may vary the most across the field. Nevertheless, other terms such as the anisoplanatism contain also low-order modes. To mitigate degeneracy issues of the best-fitting process involved by PRIME, one may use the spatial PSF diversity: the static term is constant across the field while the anisoplanatism creates a spatially variable PSF.

(ii) $\mathbf{K}_{\perp}$ is a spatial filter that includes the uncompensated spatial frequencies only by using the power-spectrum density (PSD; Jolissaint 2010; Correia & Teixeira 2014). From the PSD, we get the phase structure function (SF) of uncorrected spatial frequencies $\mathcal{D}_{\perp}$ that we normalize to $r_0 = 1$. Therefore, $\mathbf{K}_{\perp}$ is derived from r_0 and $\mathcal{D}_{\perp}$ as follows:

$$\mathbf{K}_{\perp}(\boldsymbol{\rho}/\lambda, r_0) = \exp\left(-0.5 \times r_0^{-5/3} \times \mathcal{D}_{\perp}(\boldsymbol{\rho}, r_0 = 1)\right). \quad (4)$$

(iii) $\mathbf{K}_{tt}$ is the TT spatial filter derived from the residual AO jitter. We name $w_{tt}(t)$ the time-dependent residual TT wavefront delivered by the tip-tilt sensor. $\mathbf{K}_{tt}$ is readily obtained from the tip-tilt SF $\mathcal{D}_{tt}$ as

$$\mathbf{K}_{tt}(\boldsymbol{\rho}/\lambda, g_{tt}) = \exp(-0.5 \times \mathcal{D}_{tt}(\boldsymbol{\rho}, g_{tt})), \quad (5)$$

where $\mathcal{D}_{\text{tt}}(\rho)$ is derived from w_{tt} based on

$$\mathcal{D}_{\text{tt}}(\rho, g_{\text{tt}}) = \frac{g_{\text{tt}}^2}{D^2} \times \rho \cdot (\langle w_{\text{tt}}(t) \cdot w_{\text{tt}}(t)^t \rangle_t - C_{\eta}^{\text{tt}}) \cdot \rho^t \quad (6)$$

where $\langle x \rangle_t$ is the temporal average of vector x , C_{η}^{tt} the tip-tilt noise covariance matrix and D the telescope diameter. The scalar g_{tt} is a tuning factor of the TT optical gain. Noise variance is obtained by adjusting a model of the AO loop rejection filter on temporal PSD as proposed by Jolissaint et al. (2015).

(iv) K_{dm} corresponds to the TT excluded filter calculated from the AO residual error within the AO control radius. It is calculated from $\mathcal{D}_{\text{dm}}$ the residual phase SF in the guide star direction as follows:

$$K_{\text{dm}}(\rho/\lambda, C_n^2(h), g_{\text{dm}}) = \frac{\iint_{\mathcal{P}} \mathcal{P}(\mathbf{r}) \mathcal{P}^*(\mathbf{r} + \rho) \exp(-0.5 \times \mathcal{D}_{\text{dm}}(\mathbf{r}, \rho, C_n^2(h), g_{\text{dm}})) d\mathbf{r}}{\iint_{\mathcal{P}} \mathcal{P}(\mathbf{r}) \mathcal{P}^*(\mathbf{r} + \rho) d\mathbf{r}}, \quad (7)$$

where $\mathcal{D}_{\text{dm}}$ is derived from

$$\mathcal{D}_{\text{dm}}(\mathbf{r}, \rho, g_{\text{dm}}) = g_{\text{dm}}^2 \times \mathbf{F}(\mathbf{r}) (\langle w_{\text{dm}}(t) \cdot w_{\text{dm}}(t)^t \rangle_t - C_{\eta}^{\text{dm}}) \mathbf{F}^t(\mathbf{r} + \rho), \quad (8)$$

where w_{dm} is the time-dependent residual wavefront in the DM commands space, C_{η}^{dm} is the noise covariance matrix on DM actuators, $\mathbf{F}$ the influence function matrix that converts the DM commands into a point-wise phase. Again, the scalar g_{dm} is a tuning factor of the WFS optical gain.

(v) K_{alias} accounts for the WFS aliasing effect and is derived similarly to $K_{\perp}$ by calculating the aliasing PSD following equations in (Jolissaint 2010; Correia & Teixeira 2014). From the PSD, we calculate the aliasing covariance map from the Fourier transform and apply the spatial filter formed by the DM influence functions to exclude the uncontrolled high spatial frequencies. We obtain eventually a aliasing SF $\mathcal{D}_{\text{alias}}$ normalized to $r_0 = 1$ and compute the aliasing OTF by $\exp(-0.5 \times r_0^{-5/3} \mathcal{D}_{\text{alias}}(\rho, r_0 = 1))$.

(vi) K_{aniso} refers to the anisoplanatism and relies on the calculation of the anisoplanatic SF from the $C_n^2(h)$ as presented in Beltramo-Martin et al. (2018). We use the MASS/DIMM at Mauna Kea to have an initial guess on the profile.

For on-axis PSF estimation, the vector μ contains six parameters: the seeing, two optical gains, and three Zernike modes. For off-axis observations, PRIME may retrieve the $C_n^2(h)$ over a number of layers that depends on the imager FOV and wavelength Beltramo-Martin et al. (2018). With the 10 arcsec $\times$ 10 arcsec FOV NIRC2 imager at Keck II using the K-band filter and with a 30 deg telescope zenith angle, we expect the off-axis PSF to be sensitive up to three layers (Beltramo-Martin et al. 2018), suggesting that the PSF model is parametrized with five system parameters plus three others for describing the atmosphere. When off-axis PSFs are available, PRIME uses the PSF elongation due to the anisoplanatism to constrain the altitude layers ($h > 0$ m) and retrieves the integrated seeing value from the PSF wings.

For the following, we introduce the terminology *uncalibrated* PSF-R as the PSF obtained through the reconstruction process with $g_{\text{dm}} = g_{\text{tt}} = 1$, $\mathbf{a}_z = 0$ and getting the $C_n^2(h)$ from the MASS/DIMM at Mauna Kea. The seeing is preliminary estimated using the AO telemetry by adjusting a model of the atmospheric Zernike modes variance over the reconstructed phase from the DM commands and the residual wavefront.

2.2 PSF-R optimization using best fitting

The novelty with PRIME is to be capable of retrieving the PSF-R parameters by capturing them from the focal-plane image. We use an iterative non-linear minimization to best fit the PSF-R model over observed images that must contain one or several point sources in the field at different positions. In order to account for the specificity of each reference sources in terms of photometry and astrometry, PRIME minimizes the following multiple point-sources criterion

$$\varepsilon(\mu, \mathbf{p}, \alpha) = \left\| I - \sum_{i=1}^{n_{\text{psf}}} \hat{I}(\theta_i, \mu, p_i, \alpha_i) + b \right\|_{\mathcal{L}_2(\Omega)}^2, \quad (9)$$

where I is the observed image at the sky location θ_i and $\hat{I}(\theta_i, \mu, p_i, \alpha_i)$ is the corresponding model that depends on PSF parameters μ , stellar photometry p_i , sub-pixel astrometry α_i around position θ_i , and b a mean background level. We use a $\mathcal{L}_2$ norm that integrates the residual over pixels within a square box of FOV given by Ω . We calculate the norm over at least twice the AO correction radius. The image model is then given by the following expression:

$$\hat{I}(\theta_i, \mu, p_i, \alpha_i) = p_i \times \mathcal{F}_{\Omega} \left[\widehat{\text{OTF}}_{\varepsilon}(\theta_i, \mu) \exp^{-2i\pi \times \alpha_i \rho / \lambda} \right], \quad (10)$$

where $\widehat{\text{OTF}}_{\varepsilon}(\theta_i)$ is the reconstructed OTF in direction θ_i from equation (1) that involves the AO telemetry and depends on the meta-parameters μ . The multiplication by the exponential term allows to introduce sub-pixel shifts. The operator $\mathcal{F}_{\Omega}$ applies the 2D Fourier transform and any required operations (zero-padding, cropping, interpolation) to calculate the PSF at the desired resolution, position, and pixel-scale.

Practically, we compute the PSF model over $n \times \Omega$ pixels, with $1 \leq n \leq 2$: when having multiple sources in the field, the science image is centred around an arbitrary position and some reference sources may be at the image corner. In this situation, to capture the total corner-to-corner angular frequency range, we must set $n = 2$ to mitigate periodization issues yielded by the FFT algorithm when translating the PSF from the image centre to the corner using the Fourier phasor in equation (10).

PRIME solves for the criterion in equation (9) iteratively using a Levenberg–Marquardt algorithm that derives at each iteration the PSF model gradient with respect to the adjusted parameters. The minimization process stops when reaching the stop conditions, which can be either a maximal number of iterations, or the variations on adjusted parameters or criterion, are lower than a threshold given by the user. In practice, we have found that $n_{\text{max}} = 300$, $\delta_{\text{min}} = 10^{-10}$, and $\varepsilon_{\text{min}} = 10^{-10}$ provides a satisfactory convergence for Keck II telescope NIRC2 images. Eventually, PRIME delivers astrometry and photometry of stellar sources, the post-AO PSF best-fitting model and the corresponding atmosphere and system characteristics. We summarize the PRIME architecture in Fig. 1. Also, equation (9) does include a data-based criterion only and no regularization term. To avoid the algorithm convergence to a local minimum, we pass to PRIME initial inputs that are assumed to be close to what is expected, such as the MASS/DIMM $C_n^2(h)$ and the optical gain calibration (Ragland et al. 2016) and we bound accordingly the parameters space.

The full PSF calibration process is composed by two steps: (i) direct PSF-R using the AO telemetry only, which allows to instantiate the computation of SFs in equations (4), (5), and (7). This step can take between 10 s to several minutes on a regular laptop depending on the number of layers to be retrieved, (ii) the best fitting: for a single PSF to fit, it takes roughly 10 s up to 30 s but this time increases with the number of stars. For 20 stars, this

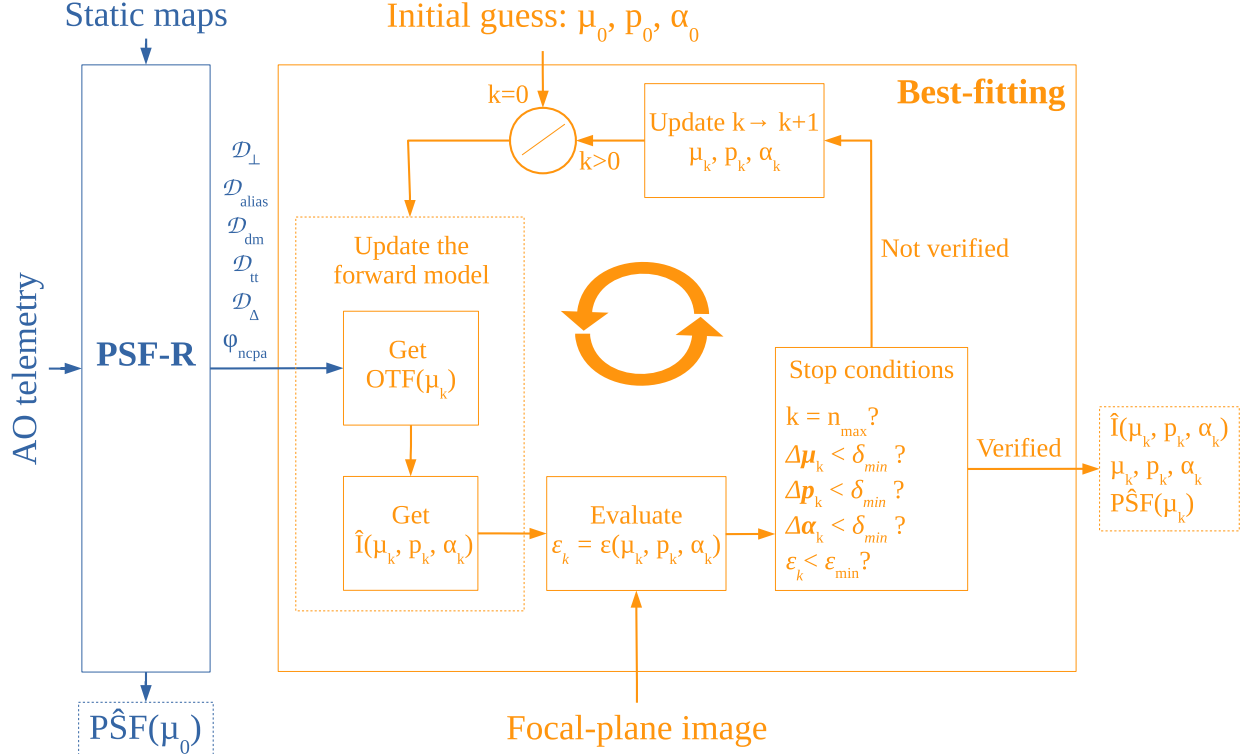


Figure 1. Block-diagram of PRIME algorithm. Increments on inputs are derived from image model gradient by the Levenberg–Marquardt algorithm. In the end of the procedure, we get the estimated PSF parameters vector $\hat{\mu}$, the stellar photometry $\hat{p}$ and astrometry $\hat{\alpha}$, the corresponding PSF and image model.

two-step full process can take 30 min in total. One of the next step of this work will concentrate in reducing this computation time as much as possible, by calculating analytically the model gradient and marginalizing the problem to retrieve the stellar parameters, which should afford a factor of two to three improvement in the computation time.

2.3 Validation from simulations

We have simulated AO-compensated H -band PSFs illustrated in Fig. 2 observed by NIRC2 and located at 10 arcsec from the guide star, which corresponds to the maximal separation between two stars observed with the NIRC2 detector in narrow-field mode with $9.94 \text{ mas pixel}^{-1}$. We have contaminated the PSF with star photon-noise, electronic noise with 38 photo-events of standard deviation, 0.08 e-/s of dark current, and $13.6 \text{ mag arcsec}^{-2}$ of sky brightness.

The simulation includes NCPA residual from on-sky calibration and anisoplanatism, with a two-layer atmospheric profile (64 percent at 180m and 36 percent at 7km) with $r_0(500\text{nm}) = 16\text{cm}$ and $L_0 = 25\text{m}$. The simulated profile is defined by compressing with the mean-weight approach (Robert et al. 2010) the median Mauna Kea profile over two layers, which is sufficient to describe the anisoplanatism at this separation (Beltramo-Martin et al. 2018). Fig. 2 shows a slight PSF elongation due to anisoplanatism, which is drowned into the noise floor when including detector noise. We have verified that the algorithm converges towards the exact solution ($\epsilon = \epsilon_{\min}$) when removing electronic and photon noise. Then, we have tuned the star magnitude and contaminated the simulated image with noise before calibrating the four parameters (layers strengths plus WFS and TT optical gains) in order to assess the algorithm robustness to noise.

We report in Fig. 3 the accuracy on the adjusted parameters as a function of the calibration star magnitude. Fig. 4 reports the corresponding errors on PSF estimates, such as the Strehl-ratio (SR), FWHM and Fraction of Variance Unexplained (FVU) that gives the overall PRIME PSF residual from the noise-free reference (Beltramo-Martin et al. 2018). SR is the most affected PSF estimate and reaches 1 percent of error for $m_H = 14$, for which we get 1 percent on the altitude seeing estimation and 5 percent on the other parameters. Also, we learn that we can tolerate 5 percent of error on optical gains to ensure a PSF modelling to better than 1 percent-level of accuracy.

Finally, the FWHM estimation is more robust to noise compared to the SR evaluation. The main explanation of this is due to the PSF halo fitting that is strongly limited by the presence of noise, which affects the SR evaluation, compared to the brighter PSF core from which the FWHM is estimated. On top of that, the altitude seeing does impact the FWHM as well by introducing more or less anisoplanatism, which permits to enhance its estimation accuracy, comparatively to the ground layer seeing measurement that suffers from the noise contamination on the PSF wings.

3 APPLICATION TO NIRC2 AT KECK II

3.1 Validation of PRIME in engineering mode

In the following, we present PRIME results applied to the near infra-red imager NIRC2 at Keck II in narrow-field mode with $9.94 \text{ mas pixel}^{-1}$ sampling. We have processed 158 images collected simultaneously with the AO telemetry over three nights, in either NGS (2013 August 1 and 2017 March 14 and 15, 80 images; Wizinowich et al. 2000) or LGS (2017 March 14, 78 images;

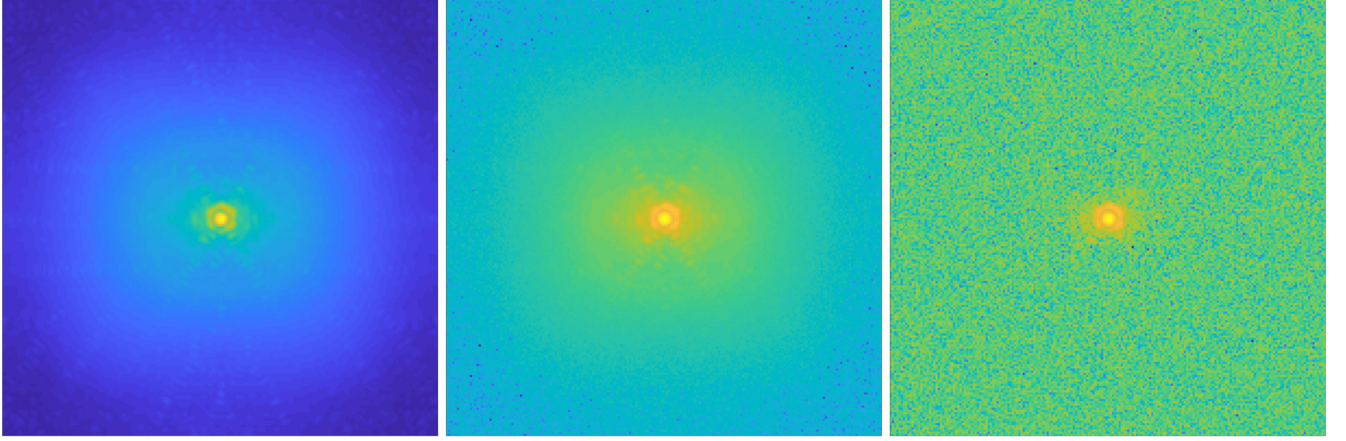


Figure 2. Simulated NIRC2 PSFs in *H*-band narrow field mode with 30 s of exposure time including electronic and photon noise. The PSF is affected by NCPA residual and is located at 10 arcsec from the guide star. The isoplanatic angle was set to $\theta_0 = 25$ arcsec and $r_0 = 16$ cm. *Left:* No noise. *Middle:* Magnitude 9. *Right:* Magnitude 14.

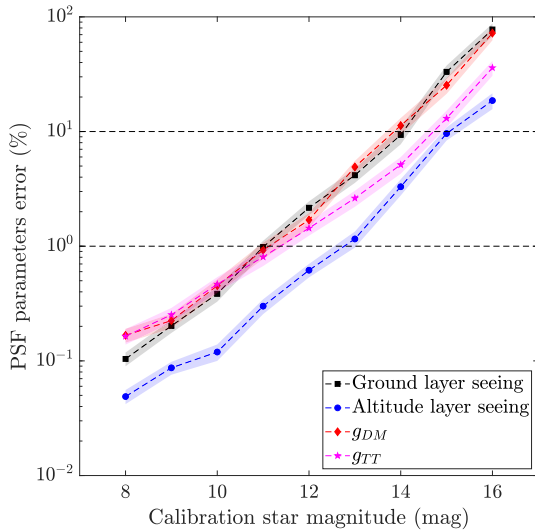


Figure 3. Accuracy on the PSF parameters estimation delivered by PRIME with respect to the calibration star magnitude with 30s of exposure time. Envelopes give the $1 - \sigma$ standard deviation.

Wizinowich et al. 2006) mode. We have systematically retrieved the seeing, optical gains, focus, and astigmatism terms by using PRIME. In LGS mode, the observations were taken with the NGS on-axis, which creates only focal anisoplanatism. This latter was not strong enough to identify a full $C_n^2(h)$ profile but does impact the SR and has been used in parallel with the PSF halo to constrain the seeing value. The sky image was cropped to 168 pixels to limit the noise propagation through the fitting process, which corresponds to four times the AO correction band.

We report in Fig. 5 the reconstructed SR and FWHM versus measured values from images in NGS ($1.65 \mu\text{m}$) and LGS ($2.12 \mu\text{m}$) modes. PRIME reaches a very good accuracy on SR and FWHM estimation of, respectively, 0.8 ± 0.5 percent and 2.8 ± 0.9 mas. Furthermore, we provide in Fig. 6 and Fig. 7 a visual inspection of sky and PRIME-reconstructed PSFs that illustrate that on top of PSF metrics, the PSF morphology is well retrieved for various conditions of performance, even with strong PSF elongation due to wind-shake (case n0070).

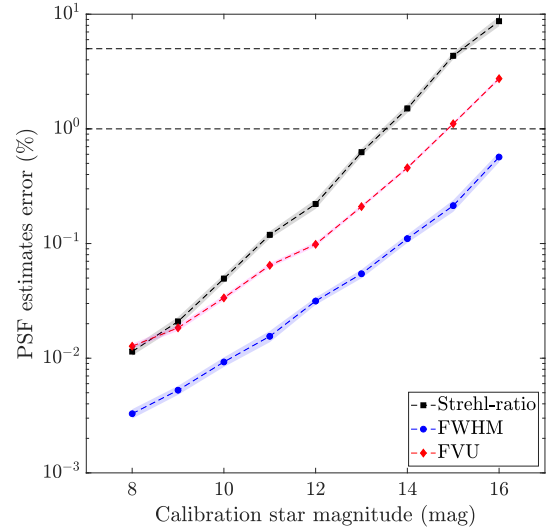


Figure 4. Accuracy on PSF estimates derived from the PRIME PSF best fit with respect to the calibration star magnitude. Envelopes give the $1 - \sigma$ standard deviation.

These results were expected in the sense that we use a best-fitting technique that must ensure that the PSF model morphology matches as well as possible the observation. Such an excellent agreement demonstrates that PSF-R accuracy is only limited by the determination of key parameters: seeing optical gains and static phase, this latter especially when the performance regime is high ($\text{SR} > 40$ per cent).

3.2 Validation of PRIME on binary stars

To pursue the validation of PRIME capabilities, we have also processed images of binary 2MASS00063057+2711500 and 2MASS00535794+2535445 acquired with NIRC2 in 2013 August 1. NIRC2 was configured in narrow-field band, and five observations were done using the FeII filter ($1.6455 \mu\text{m}$) over 50 s exposure each. We have measured the stability of the photometry and astrometry measurements by using uncalibrated PSF-R, PRIME, and a Moffat model. We present in Fig. 8 a 2D visual comparison of best-fitting

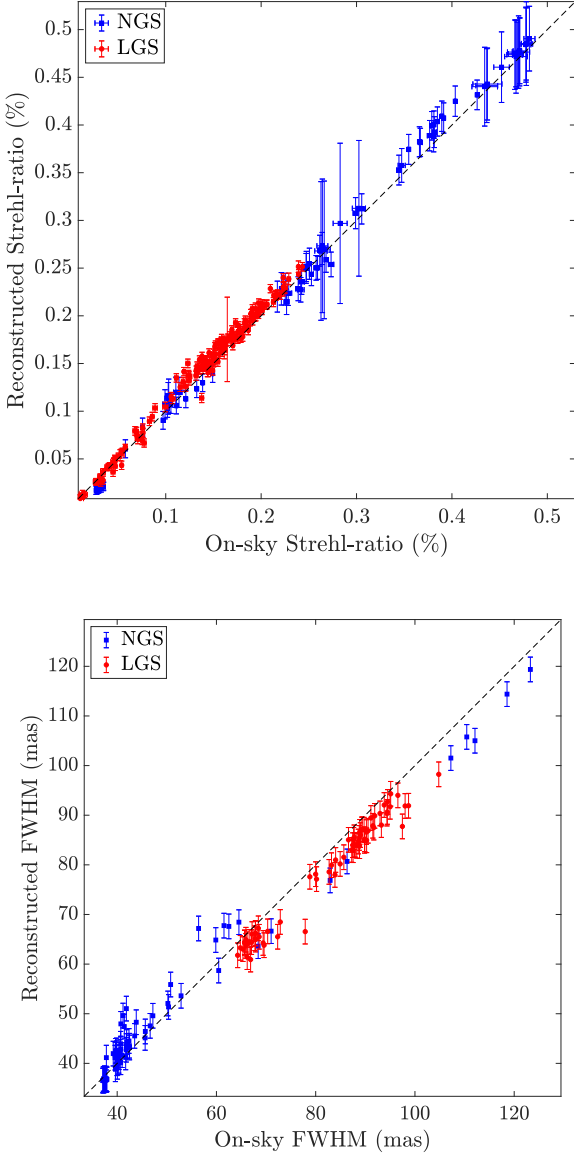


Figure 5. *Top:* Reconstructed SR. *Bottom:* FWHM versus measurements over images obtained through PRIME in NGS (blue) and LGS (red) modes. Dashed line is the $y = x$ line.

models as well as azimuthal profile averaged out over the five frames. Present results confirm that PRIME can estimate jointly PSF and stellar parameters efficiently. A single Moffat function is not accurate enough to reproduce the entire PSF shape, and the uncalibrated PSF-R is not sufficiently accurate to reach the same residual level as PRIME. We particularly notice that PRIME has played on the seeing value to adjust the PSF wing level. We report in Tab. 1. the residual error obtained with each of the three models that confirms quantitatively that PRIME improves estimates precision by 2 0% and 40% compared to respectively a forward PSF-R and Moffat model.

Results illustrate that PRIME improves the precision on stellar parameter estimates but does potentially increase the noise propagation though the iterative minimization loop. To check this point, we have artificially played on the object digital intensity to change its apparent magnitude using NIRC2 photometric specifications and simulate the observation we would have with a different signal-

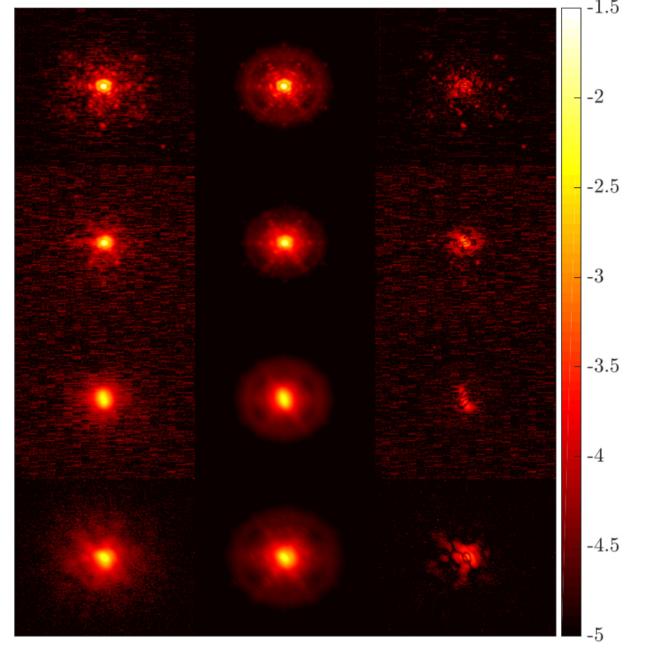


Figure 6. Visual inspection of PRIME reconstruction results. *Left:* On-sky PSFs. *Middle:* PRIME best fit. *Right:* Reconstruction residual. First row: n0004 case in NGS mode in H band (2013 August 1). Second row: n0089 case, NGS mode, H band (2013 August 1). Third row: n0070 case, NGS mode, H band (2013 August 1). Fourth row: n0086 case, LGS modes, K band (2017 March 14).

to-noise ratio (S/N). The goal of our approach is to capture how estimates precision degrade with respect to the star magnitude and the PSF model. Note that we do not touch the WFS noise, in a way we conserve the same PSF model that is calibrated for this specific observation.

We illustrate in Fig. 9 the photometry and astrometry precision with respect to the artificial object magnitude. This plot shows two distinct regimes below and beyond 14 mag that, respectively, correspond to PSF model and noise limitation regimes. For stars with $m_H > 14$ mag, the noise propagation through the fitting process dominates the estimation precision, while for stars with $m_H \leq 14$ mag for a given object magnitude, we improve the precision by a factor of 2 and 1.5 on, respectively, the photometry and astrometry; or in other words, we get the same precision for objects 1–2 mag fainter. Also, because PRIME does estimate PSF parameters in addition to photometry and astrometry, we were expecting more sensitivity to noise and see the metrics precision getting worse faster than the uncalibrated PSF-R model, which does not occur. Finally, we understand that the present implementation of PRIME is efficient when feeding it with image of stars of magnitude $m_H \leq 14$ mag stars with 50 s of exposure time, which is accessible in the GC (Yelda et al. 2010) and makes it feasible to deploy PRIME on such a science case.

4 APPLICATION OF PRIME BY-PRODUCTS

4.1 PSF-R calibration

PRIME differs from PSF-R by handling focal-plane images as well as AO telemetry to estimate the PSF. This highlights that PRIME can be operated only when bright stars are present in the science image. If not, one must rely on forward PSF-R to provide such PSF

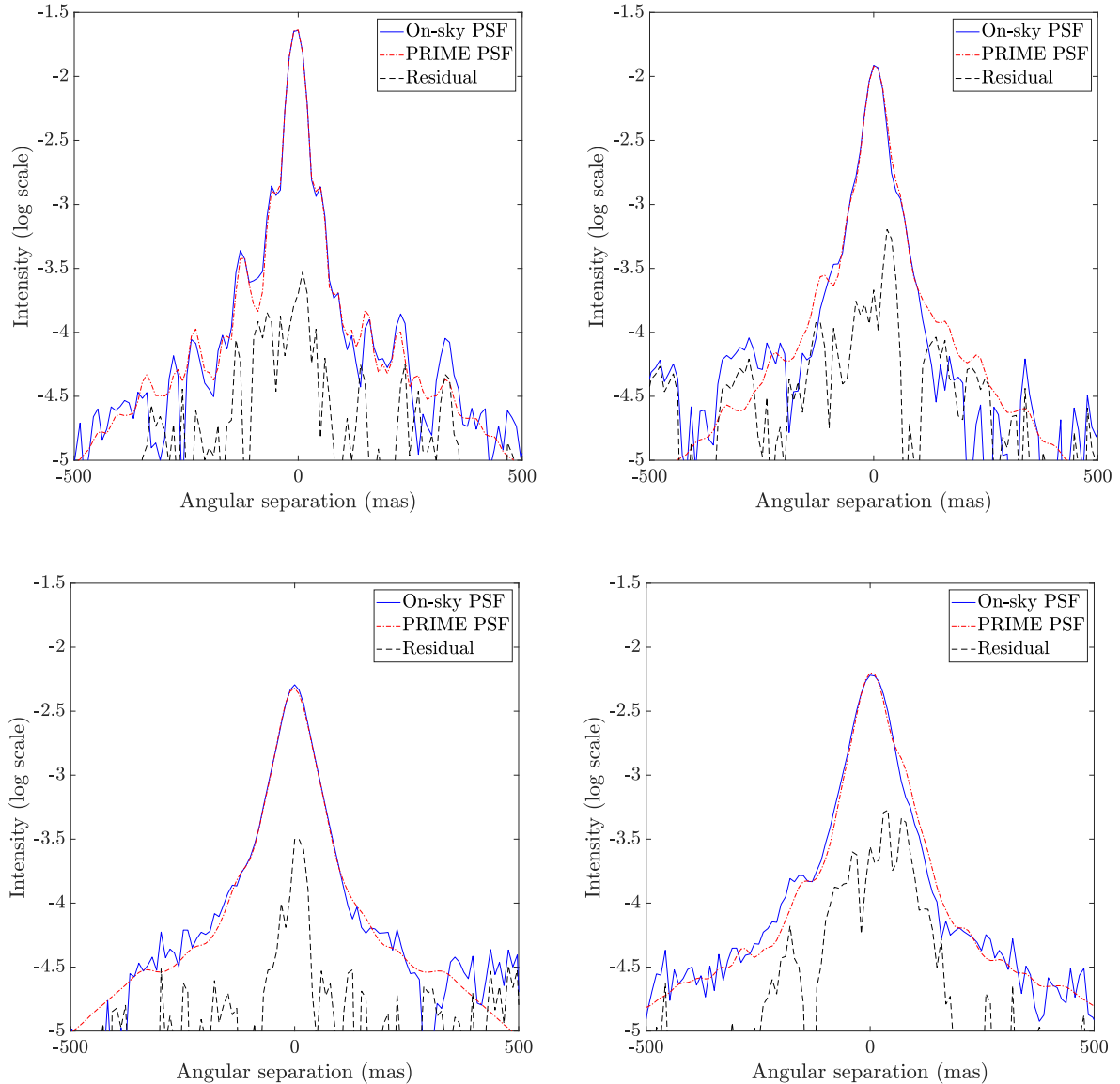


Figure 7. Cuts of sky PSF compared to PRIME best fit. *Top left:* n0004 case in NGS mode in H band (2013 August 1). *Top right:* n0089 case, NGS mode, H band (2013 August 1). *Bottom left:* n0070 case, NGS mode, H band (2013 August 1). *Bottom right:* n0086 case, LGS modes, K band (2017 March 14).

estimate. However, PRIME offers the possibility to optimize the PSF-R process by calibrating the critical parameters with respect to the contextual information that comes with the observation. In other words, we propose to infer statistical relationships using Machine Learning process between the PSF parameters and the available data. To do so, we need archive data on which we deploy PRIME and measure the seeing [or $C_n^2(h)$] and system characteristics, which are then correlated with any relevant information we may have such as the AO-telemetry seeing in the present analysis.

We have split our sample of 158 data sets into a training data set of 80 NGS data, another training set of 38 LGS data and a test set of 40 LGS PSFs. Because atmosphere characteristics do not depend on the AO configuration, the seeing was identified on both training sets to enlarge the number of training data sets. On the contrary, optical gains and static modes were measured on the LGS training set only in order to ensure that the system hardware implementation is the same for both training and learning sets.

We have assumed that the seeing and optical gains were varying linearly with the AO telemetry seeing; therefore, it is enough to deploy standard linear regression tools. We have also assumed that the Zernike term were constant, i.e. we took the median of calibrated values on the training set. All the errors bars in the following are obtained from the Levenberg–Marquardt best-fitting procedure that delivers the estimates precision at 3σ from the PSF model gradients calculated empirically.

We report in Fig. 10 the seeing retrieved with PRIME on both NGS and LGS images compared to the AO telemetry-based estimation. It highlights not only a correlation but also a very large discrepancy. This is probably due to the seeing estimation method that relies on the Zernike variance decomposition of pseudo open-loop slopes that include unmodelled aliasing effect. Moreover, the seeing measurement may encounter some issues with LGS case for which low-order modes can be introduced unwillingly on top of the atmospheric modes, as we will see it in Section 4.2. One of the

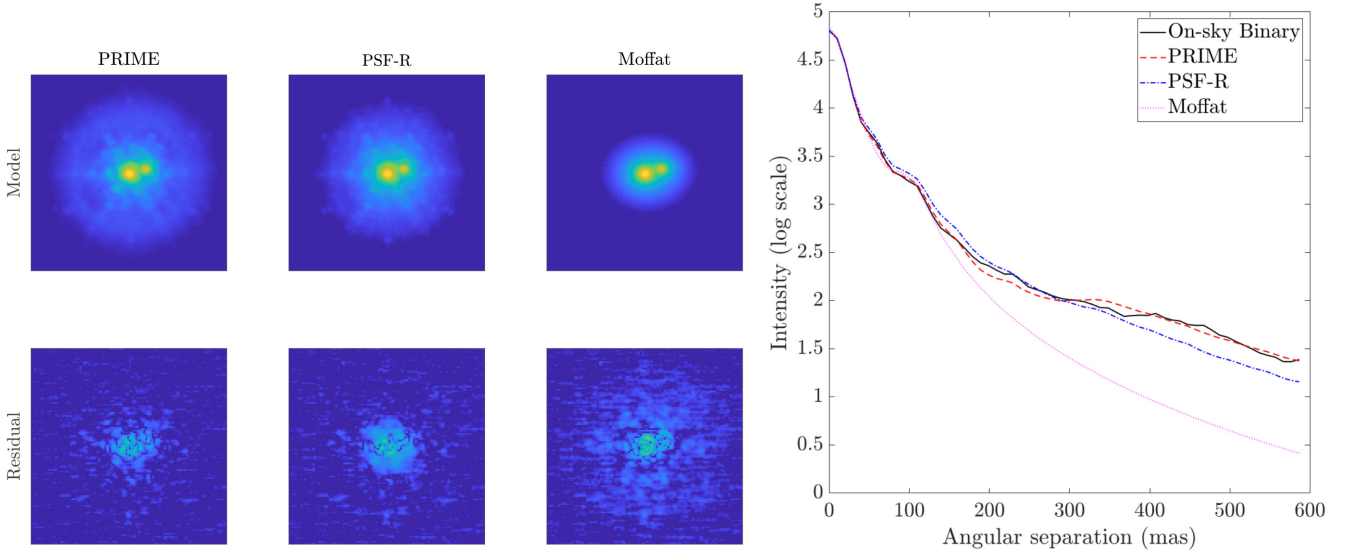


Figure 8. Illustration of the analysis on 2MASS00535794+2535445. *Left-hand panel:* Top images are, respectively, PRIME, uncalibrated PSF-R, and Moffat fitting; bottom figures give the fitting residual over sky images. *Right-hand panel:* Azimuthal average of the mean binary image over the five acquisition with overplotting of best fit obtained with PRIME, PSF-R, and a PSF Moffat model.

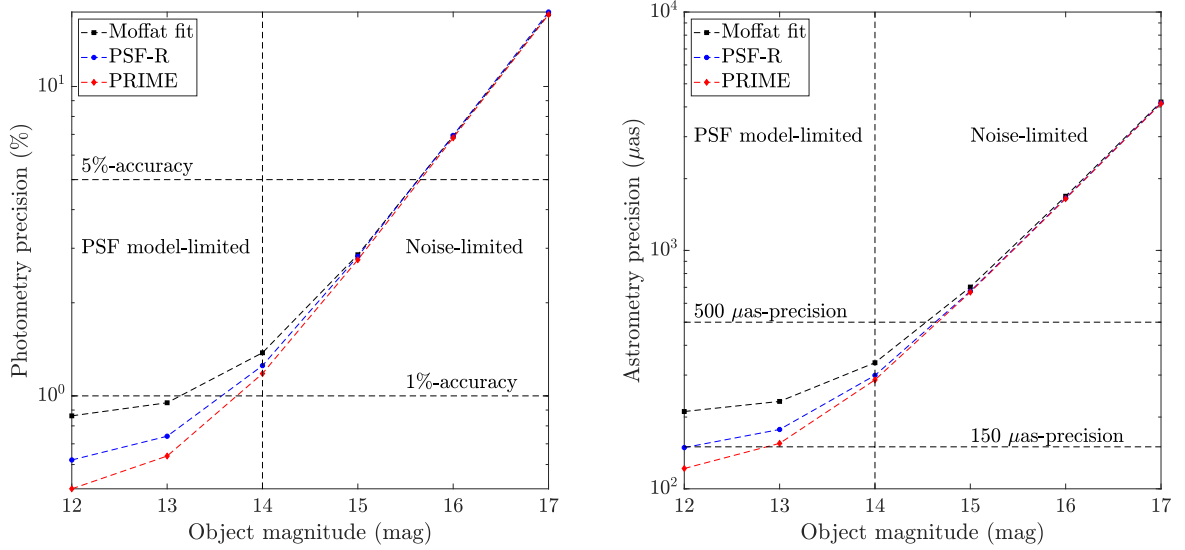


Figure 9. *Top:* Photometry. *Bottom:* Astrometry precision on binary object regarding the object magnitude (50 s of exposure time) and the PSF model. Results are averaged out over the five frames of each of the two observed binary.

next step will be to update the seeing estimator by using approaches proposed by Andrade et al. (2019) and Jolissaint et al. (2018).

However, we can highlight which gain we can expect by calibrating the seeing at zenith as $0.45r_0^{\text{tel}} + 6.7$. Also, we observed that the HO and TT gains were pretty constant over the learning set with $g_{\text{DM}} = 1.45 \pm 0.23$ and $g_{\text{TT}} = 1.31 \pm 0.11$. Retrieved Zernike mode coefficients were quite dispersed and we consider their mean values over the learning set, which are at $a_4 = 63$ nm, $a_5 = 118$ nm, and $a_6 = 79$ nm.

The next step of this analysis has consisted in comparing PSF-fitting photometry to aperture photometry, which gives the reference photometry value, over the test data set. We have compared three different PSF models: (i) *uncalibrated PSF-R* as the PSF-R using a priori parameters ($g_{\text{DM}} = g_{\text{DM}} = 1$, $a_4 = a_5 = a_6 = 0$, seeing given

by the AO telemetry); (ii) *calibrated PSF-R* as the PSF-R using the parameters calibration on the learning data set; and (iii) *PRIME* as the PSF-R optimized for each test data sample using the best fitting.

We report in Fig. 11 the photometric accuracy with respect to the seeing and we report in Table 2 the corresponding error statistics in terms of bias and standard deviation. FWHM results are very similar between the three methods, but slightly better with PRIME and the calibrated PSF-R compared to the uncalibrated PSF-R, advocating that the PSF FWHM is already well estimated from the AO telemetry only. Results highlight that the PSF-R calibration corrects for the bias on estimates and increases the precision by 50 per cent compared to the uncalibrated PSF-R. If point sources are available in the field to optimize the PSF-R using PRIME, we can expect a photometric accuracy at the level of 0.008 ± 0.004 mag

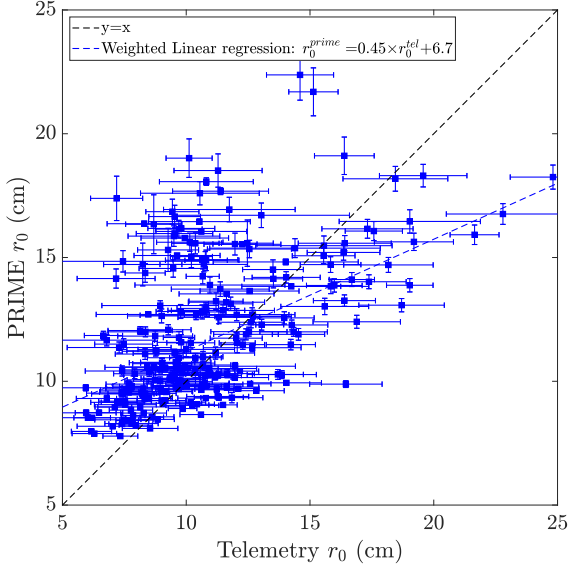


Figure 10. Seeing value retrieved by PRIME as function of the seeing estimated from the AO telemetry.

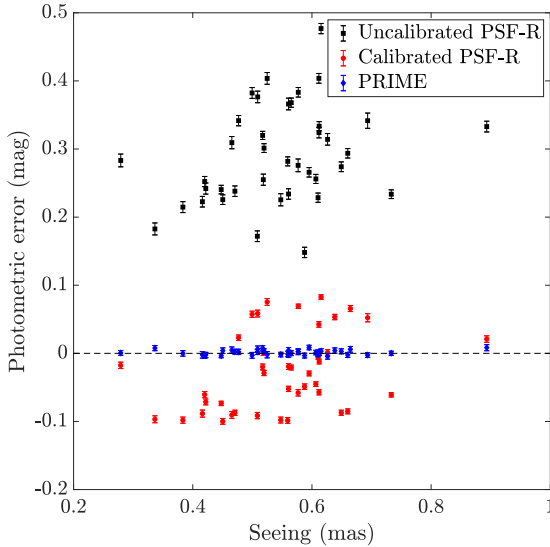


Figure 11. Photometry error versus the seeing for the 40 LGS test samples. The figure compares the uncalibrated PSF-R (optical gains set to one and use of the DIMM seeing), calibrated PSF-R (use of inferred parameters evolution laws), or PRIME (focal-plane best fit).

that degrades to 0.06 ± 0.004 mag and 0.3 ± 0.008 mag by using, respectively, the calibrated and uncalibrated PSF-R.

Such results illustrates that PRIME can be deployed to calibrate the PSF-R computation in case the PSF best fitting is compromised and ensures that PSF-R can deliver 6 percent of photometry accuracy. Such results will be improved by increasing the size of the learning data set and choosing other descriptors than the AO telemetry seeing, such as the full residual wavefront and the telescope configuration.

Table 1. Residual integrated over the image (FVU) and statistics on estimation stability of the binary separation and differential photometry regarding the PSF model. Values are averaged over the two objects.

	FVU (per cent)	Δr (μ as)	Δm (per cent)
Moffat	1.67 ± 0.34	473 ± 18	1.4 ± 0.1
PSF-R	0.77 ± 0.16	333 ± 63	1.0 ± 0.1
PRIME	0.49 ± 0.05	263 ± 31	0.8 ± 0.05

Table 2. Statistics on SR and photometric error reconstruction using either the uncalibrated PSF-R (optical gains set to one and use of the DIMM seeing), calibrated PSF-R (use of inferred parameters evolution laws), or PRIME (focal-plane best fit) over the LGS test images.

	Δ SR (per cent)		Δ mag (mag)	
	Mean	RMS	Mean	RMS
Uncalibrated PSF-R	8	3.2	0.22	0.12
Calibrated PSF-R	-0.8	2.0	-0.03	0.06
PRIME	0.7	0.3	0.0009	0.004

4.2 Image-assisted wavefront error breakdown

PSF-R is a generalization of the AO wavefront error budget: we do calculate covariance matrices of errors instead of variance terms only. From the phase SFs that appear in the PSF model described in Section 2.1, one may derive the wavefront error budget. With PRIME, we basically scale these phase SFs regarding how they do impact the PSF morphology, in a way the absolute value of wavefront error terms are eventually updated through the best-fitting process. The use of the focal-plane image adds a constrain on the SR very close to the image SR. According to equation (1), we derive the following error breakdown:

$$\sigma_{\varepsilon}^2 = \sigma_{\text{nepa}}^2 + \sigma_{\text{stat}}^2 + \sigma_{\text{fit}}^2 + \sigma_{\text{dm}}^2 + \sigma_{\text{noise}}^2 + \sigma_{\text{alias}}^2 + \sigma_{\text{jitter}}^2 + \sigma_{\text{tilt-noise}}^2 + \sigma_{\text{aniso}}^2, \quad (11)$$

where

(i) σ_{nepa}^2 is the residual error from the NCPA calibration. It is derived from the rms value of the calibrated map.

(ii) σ_{stat}^2 is the variance of the additional low-order Zernike modes adjusted by PRIME.

(iii) σ_{fit}^2 is the fitting error that corresponds to the uncompensated high-spatial frequencies and derived from the adjusted seeing estimation from PRIME by integrating the fitting PSD over the frequency ranges contained into the PSF angular support.

(iv) σ_{dm}^2 is the DM bandwidth error obtained from the trace of the matrix $\hat{g}_{\text{dm}}^2 \times (\langle w_{\text{dm}}(t) \cdot w_{\text{dm}}^t(t) \rangle - C_{\eta}^{\text{dm}})$ that appears in the calculation of OTF_{dm} in equation (7). Moreover, we subtract quadratically the noise error σ_{noise}^2 that includes the noise propagation through the AO loop, which differs from the trace of C_{η}^{dm} that includes only the WFS noise projected on to the DM actuators domain. It gives the AO correction residual excluded from high-spatial frequencies, tip-tilt modes and measurements errors, which corresponds to the servo-lag error.

(v) σ_{noise}^2 is the filtered WFS noise error that affects the PSF and derived from the trace of $p_n \times C_{\eta}^{\text{dm}}$, where p_n is the noise propagating factor calculated by integrating the noise transfer function.

(vi) σ_{alias}^2 is the filtered WFS aliasing error derived by integrating the aliasing PSD, scaled regarding the seeing value obtained by PRIME, over the frequency range covered by the AO system.

Table 3. Wavefront error breakdown obtained with PRIME. The seeing condition was similar with $r_0 = 10$ cm at zenith and 500 nm.

	LGS case	NGS case
K-band Strehl	19.8 per cent	41.8 per cent
Wavefront error	430	316
NCPA calibration	167	167
Low-order modes	161	38
Atmospheric fitting	141	150
Servo-lag error	149	141
WFS noise	58	85
WFS aliasing	38	66
Anisoplanatism	162	0
Tip-tilt bandwidth	237	118
Tip-tilt noise	42	46

(vii) σ_{jitter}^2 is the jitter error given by the trace of the matrix $\hat{g}_{\text{tt}}^2 \times (\langle w_{\text{tt}}(t) \cdot w_{\text{tt}}^t(t) \rangle - C_{\eta}^{\text{tt}})$ that gather the residual tip-tilt error. As similarly as for σ_{dm}^2 , we do subtract the loop-filtered tip-tilt WFS noise $\sigma_{\text{tilt-noise}}^2$.

(viii) $\sigma_{\text{tilt-noise}}^2$ is obtained in the same way as σ_{noise}^2 . The noise propagation is obtained using the tip-tilt loop noise transfer function that generally differs from the high-order loop in terms of gain and frequency sampling.

(ix) $\sigma_{\text{anisoplanatism}}^2$ refers to the anisoplanatism composed by three different species: angular (off-axis operations), focal, and tilt (LGS operations) anisoplanatism errors. For the present analysis, LGS cases were contaminated by focal anisoplanatism only and the NGS was always on-axis. Errors are derived from the filter K_{ani} after retrieving the seeing and the $C_n^2(h)$ (for off-axis operations) with PRIME.

We present in Table 3 results obtained from equation (11) on two different data set obtained on 2017 March 14 in either LGS or NGS mode. The seeing was very similar for those two cases with $r_0 = 10$ cm, which explain why the fitting errors were barely identical. The aliasing and DM measurement contributions take into account the AO-loop filtering, and variations from LGS to NGS cases are mostly explained by a different loop configuration that was running at 437 Hz with a gain of 0.5 and 1 kHz with a gain of 0.25, respectively, in LGS and NGS modes. If we approximate the DM control as a first-order filter, the system bandwidth and, consequently, the bandwidth error as well scale with the product gain $\times$ frequency sampling, which is therefore maintained at the same level in both cases and justify why the DM bandwidth errors are very similar.

We notice that a part of the performance degradation using the LGS is due to the tip-tilt bandwidth increasing, mostly because the tip-tilt loop was running at 125 Hz instead of 437 Hz in NGS mode and with a loop gain of 0.1 instead of 0.3. The WFS tip-tilt noise reached 200 nm in LGS mode versus 100 nm in NGS mode because of the tip-tilt star brightness. Both of them reduced to 45 nm after propagation from the tip-tilt loop. This can be verified by considering that the noise propagation factor through a first-order filter is $g/(2 - g)$, which gives $\sqrt{0.1/1.9} \times 200 \simeq 45$ nm in LGS mode and $\sqrt{0.3/1.7} \times 100 \simeq 42$ nm in NGS. Note that these values are slightly different from the ones displayed in Table 3 due to the fact that the Keck AO loop does not behave exactly as a first-order filter (van Dam et al. 2006). What we learn a posteriori is that we could have boosted the tip-tilt loop gain to optimize the overall AO performance, which would have the effect of increasing the tip-tilt noise but diminishing the tip-tilt bandwidth.

Another reason that degrades the LGS performance is the focal anisoplanatism that accounts for $162^2/430^2 = 38$ per cent of the total AO error. By using the equation (16) in Hardy (1998),

$$\sigma_{\text{anisoplanatism}}^2 = D^{5/3} \left[0.5 H^{-5/3} \sec(\gamma)^{5/3} \int_0^H C_n^2(h) h^{5/3} dh - 0.452 H^{-2} \sec(\gamma)^{5/3} \int_0^H C_n^2(h) h^2 dh \right], \quad (12)$$

where H is the sodium layer height (87.5 km) and γ the telescope zenith angle. We retrieve $\sigma_{\text{anisoplanatism}}^2 = 158$ nm, which is very close to the value we measure from the integral of the anisoplanatism transfer function K_{ani} presented in Table 3. Similarly, the fitting error we obtain from the formula $\sqrt{0.23 \times (0.5625/r_0)^{5/3}}$ is 150 nm, which compares very well to what we find from the PSD, although we probably slightly underestimate the fitting error because the spatial frequency range truncation.

Finally, we have also detected presence of low-order modes that do only occur in LGS mode and are composed by up to 120 nm of focus and 100 nm of astigmatism. Wavefront control using LGS may suffer by focus mis-estimation because the LGS height variations that creates focus modes independent to the phase atmospheric distortion. From Wizinowich (2012), this error is expected to be at the level of 70 nm for median seeing ($r_0 = 16$ nm) and sodium profile conditions. Considering we have $r_0 = 10$ cm, we should therefore find error about $70 \times (0.16/0.1)^{5/3} = 105$ nm, which is close to what we find with PRIME. It still remains 100 nm of unexpected astigmatism. Nevertheless, we have the certainty that this astigmatism is introduced by the LGS control when comparing to the static modes retrieved in NGS mode, which are meaningless in terms of wavefront error. Keck II is equipped with a low-bandwidth wavefront sensor (LBWFS; Wizinowich et al. 2006) that is supposed to capture these low-order modes and update the AO loop at a low temporal frame rate to correct for them. The LBWFS telemetry is not yet available into the Keck Telemetry Recording System (TRS), but additional investigation will be led to verify whether the LBWFS sees this astigmatism or not. To summarize, PRIME offers another look at the AO error breakdown and has the potentiality to be used as an AO diagnosis facility.

4.3 Relative astrometry and photometry

As we have already introduced, PRIME can be used to estimate stellar parameters of multiple sources as well. So far, we do not think that PRIME is mature enough to replace any existing pipeline for sources characterization, we preferably aim at coupling PRIME with existing pipelines, such as AIROPA (Witzel et al. 2016; Ciurlo et al. 2018), to increase the stellar estimates accuracy. We plan to calibrate the PSF model over extracted point-source images across the field and generate a PSF grid to account for spatial (spectral as well if necessary) variations, which will be handled by AIROPA to obtain stellar parameters.

To validate this process, we have applied PRIME on 82 observations taken with NGS AO on 2017 March 15 of the triple system Gl 569 formed by a low-mass binary component Gl 569B located 4.9 arcsec from the primary Gl 569A. The AO loop was closed on Gl 569A and unsaturated images of all three components were obtained in the K cont. filter ($\lambda_{\text{eff}} = 2.2705 \mu\text{m}$) over 20 coadds of 0.2 s exposure each. Thanks to the presence of the third companion Gl 569A, we have extracted out the PSF from the scientific image and use it to measure separation and differential photometry by best fitting.

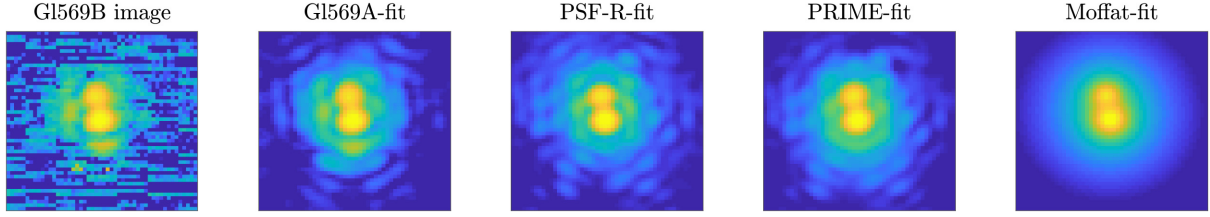


Figure 12. From left to right: (1) *H*-band NIRC2 image of Gl 569B acquired over 20 coadds of 0.2 s integration each, (2) best fit using the on-axis NGS image, (3) best fit using the uncalibrated PSF-R model, (4) best fit using the optimized PSF-R model with PRIME, and (5) best fit using an asymmetric Moffat PSF model.

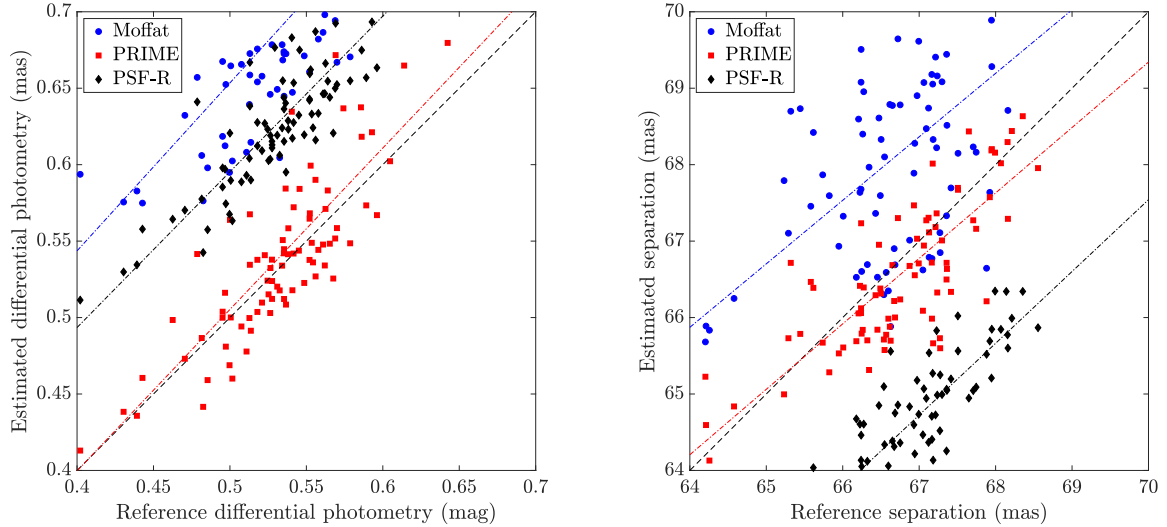


Figure 13. Estimated differential photometry (top) and binary separation (bottom) obtained by modelling the PSF with a Moffat, uncalibrated PSF-R and PRIME with respect to the Gl 569A-based retrieval. Each sample corresponds to one of the 82 acquisitions. The dash line corresponds to $y = x$ and the dashed-dotted lines refer to linear regression for each model.

Table 4. Statistics on estimation of the binary separation and differential photometry regarding the PSF model. Error bars are obtained from an average over 82 successive acquisitions.

PSF model	Separation (mas)	$m_A - m_B$ (mag)
Gl 569A	66.76 ± 0.94	0.532 ± 0.041
PRIME	66.73 ± 1.02	0.538 ± 0.048
PSF-R	64.50 ± 1.00	0.628 ± 0.051
Moffat	68.17 ± 1.35	0.689 ± 0.061

Then we have repeated the same process by using alternative PSF models: (i) uncalibrated PSF-R, (ii) asymmetric Moffat model, and (iii) PSF-R model optimized by best fitting the image of Gl 569A. For each of those three models, we have also included the angular anisoplanatism by relying on MASS/DIMM $C_n^2(h)$ values. Regarding the short separation of the binary from Gl 569A, the imaging wavelength (*K* filter) and the telescope zenith angle (almost zenith), the anisoplanatism was not sufficiently strong to impact substantially the PSF at the binary location. On top of that, the S/N on binary frames is not good enough to reliably retrieve $C_n^2(h)$. We have checked that not including the anisoplanatism model has no noticeable impact on results, i.e. changes on results are lower than error bars, in the sense that using the MASS/DIMM measurements suits perfectly to this situation in order to mitigate any potential measurement bias.

We report in Fig. 12 results of the fitting process for a particular acquisition compared to the original image of Gl 569B. The figures highlight that the binary observation is very well reproduced when relying on the extracted PSF; regarding the position of Gl 569A to the binary and the imaging wavelength, the anisoplanatism effect is extremely weak and can be neglected, in a way that the PSF extraction does not suffer substantially from any PSF model error. The reconstruction stands very close to the real PSF but miss its asymmetric shape, which is potentially introduced by a particular pattern of co-phasing errors of the primary mirror segments that produces low-order modes aberrations (Ragland 2018). PRIME succeeds partially restoring the asymmetric shape, while the Moffat model is not accurate enough to mimic the complex PSF structure.

We report in Table 4 estimation results averaged over the 82 acquisitions that confirm the visual inspection on Fig. 12: by adjusting the PSF-R model over Gl 569A, PRIME allows to catch up reconstruction errors to achieve stellar parameters retrieval similar to the PSF extraction approach technique. This is also confirmed by Fig. 13 that illustrates that each method provides estimates highly correlated to reference values, but biased except when using PRIME. We have also tested to retrieve conjointly the PSF and binary parameters by best fitting directly the binary image itself with a two-PSF model. We reached very similar results compared to the scenario PSF model calibration on Gl 569A plus photometry/astrometry estimation by PSF fitting. Differences are within the uncertainty bars, although Gl 569B is 3.7 mag fainter

than Gl 569A, which suggests that the joint estimation should be more efficient than the sole PSF calibration at equivalent S/N. This also points out that providing multiple sources PSF to PRIME allows effectively to strengthen the best-fitting process and mitigate the noise contamination, which suits ideally to PSF estimation in crowded fields.

5 CONCLUSION

We have introduced PRIME as a new approach that couples AO telemetry and focal-plane images to provide (i) atmospheric and system parameters such as the $C_n^2(h)$ profile, system optical gains low-order residual error, (ii) an accurate error breakdown and thereupon PSF model across field locations and wavelengths, and (iii) photometry and astrometry estimates on fields with multiple sources where a non-linear least-squares PSF fitting routine is employed. In addition, statistical analysis with PRIME provides useful information and trends of parameter variation with respect to observing conditions thus being valuable to optimize runtime AO performance and to calibrate a priori parameters for later PSF-R.

We have deployed PRIME on Keck II on-sky PSF in engineering mode when guiding either on an NGS or on an LGS axis. We have demonstrated that only few parameters must be estimated carefully (seeing, optical gains plus additional focus and astigmatism terms) to reach 0.8 ± 0.5 per cent of accuracy on SR, 2.8 ± 0.9 mas on FWHM, and 0.008 ± 0.004 mag on photometry. Over a sub-sample of data, we have calibrated the PSF-R model accordingly to the AO telemetry-based seeing that has been used to calculate the PSF from the direct model. We have shown that this calibration permits to decrease the photometry error from 0.3 ± 0.008 mag down to 0.06 ± 0.004 mag.

We have presented an application of PRIME to constrain the a posteriori the AO wavefront error breakdown absolute value with respect to the impact of each error term on the PSF morphology. We have particularly highlighted that LGS data acquired in 2017 March 14 were suffering from 100 nm of astigmatism on top of what was seen in the AO telemetry.

We used tight-binary observations and analysed, on the one hand, the robustness to noise and on the other the PSF model accuracy. We have illustrated on NIRC2 binary images that PRIME is sufficiently robust to noise to retain photometry and astrometry precision below 0.005 mag and 100 μ as on a $m_H = 14$ mag object, whereas a loss is to expect when using Moffat models or pre-calibrated PSF parameters. This is indicative of the usefulness of estimating PSFs based on the actual observations (images and telemetry) to avoid errors to creep in and degrade seek-after observable. Finally, we have also highlighted that PRIME can perform a PSF calibration on the triple system Gl569BAB that provides a separation of 66.73 ± 1.02 mas and a differential photometry of 0.538 ± 0.048 mag, compared to the reference values obtained with the extracted PSF that are 66.76 ± 0.94 mas and 0.532 ± 0.041 mag. The next step of this work will consist in coupling PRIME-delivered PSFs with existing source extraction software to increase the accuracy of astrometry and photometry measurements on crowded fields observations. Moreover, we will deploy the PSF-based AO diagnostic to improve a posteriori the AO running operations.

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